Machine Talk: How Verbal Embodiment in Conversational AI Shapes Consumer–Brand Relationships

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This research shows that AI-based conversational interfaces can have a profound impact on consumer–brand relationships. We develop a conceptual model of verbal embodiment in technology-mediated communication that integrates three key properties of human-to-human dialogue—(1) turn-taking (i.e., alternating contributions by the two parties), (2) turn initiation (i.e., the act of initiating the next turn in a sequence), and (3) grounding between turns (i.e., acknowledging the other party's contribution by restating or rephrasing it). These fundamental conversational properties systematically shape consumers' perception of an AI-based conversational interface, their perception of the brand that the interface represents, and their behavior in connection with that brand. Converging evidence from four studies shows that these dialogue properties enhance the perceived humanness of the interface, which in turn promotes more intimate consumer–brand relationships and more favorable behavioral brand outcomes (greater recommendation acceptance, willingness to pay a price premium, brand advocacy, and brand loyalty). Moreover, we show that these effects are reduced in contexts requiring less mutual understanding between the consumer and the brand. This research highlights how fundamental principles of human-to-human communication can be harnessed to design more intimate consumer–brand interactions in an increasingly AI-driven marketplace.

Keywords: AI, conversational interfaces, chatbots, natural language processing, brand intimacy, consumer–brand relationships, technology-mediated branding

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Advances in natural language processing and the increased adoption of messaging platforms are transforming consumer–brand experiences. The use of AI-based conversational interfaces as a novel interaction paradigm between consumers and brands has been characterized as the "next operating system in commerce" (Suri, Elia, and van Hillegersberg 2017) and is transforming how consumers search and shop for products. By deploying such conversational interfaces (also commonly referred to as "chatbots" or "conversational AI"), brands can enable consumers to engage in two-way conversations to obtain product information, assistance with product selection, and after-sales support (Clark 2017). For instance, retailers such as Sephora have adopted chatbots to provide virtual product consulting, shopping assistance, and scheduling support for in-store appointments (Taylor 2016). Yet, the implementation of AI-based conversational interfaces varies substantially (see web appendix 1 for examples). While consumers often appreciate the speed and efficiency of such interfaces, they might also feel apprehensive about chatbots replacing human-to-human interactions (Crolic et al. 2021; Kaneshige and Hong 2018). Indeed, consumers tend to perceive AI (and, by extension, the organization that deploys it) as mechanical and cold (Granulo, Fuchs, and Puntoni 2019; Longoni, Bonezzi, and Morewedge 2019). However, little is known about the downstream consequences for a brand that a conversational AI represents and how the specific conversational design features of such interfaces might impact consumer–brand relationships.

The primary focus of prior research on AI-based conversational interfaces was on design- and technology-related considerations such as optimizing the visual appeal and usability of these interfaces (Landis 2014; Lokman and Zain 2010); the role of speed and efficiency in promoting user acceptance (Comendador et al. 2015; Go and Sundar 2019; Hildebrand and Bergner 2021; Lee, Lee, and Sah 2019; Zarouali et al. 2018); the use of anthropomorphic elements (Araujo 2018; Crolic et al. 2021); or how automated interfaces might learn through interactions with a user (Hildebrand et al. 2020; Serban et al. 2017). The impact of conversational interfaces on consumer–brand interactions and their influence on brand experience are not well understood. The limited prior work on this phenomenon has primarily focused on the role of avatars, or other visual characteristics of embodiment, on consumer perception (Bertacchini, Bilotta, and Pantano 2017; Holzwarth, Janiszewski, and Neumann 2006; Jin and Sung 2010; Wang et al. 2007).

The current research introduces a theoretical framework that characterizes how key conversational properties of an AI-based conversational interface initially affect consumers' perception of that interface, shape a response that influences the relationship consumers develop with the brand, and ultimately have important downstream consequences on consumer choice. Our framework builds on the idea of consumer–brand relationships as an active social process (Brick and Fournier 2017; Fournier 1998; MacInnis and Folkes 2017; Reis and Shaver 1988; Thomson 2006). We integrate prior work on consumer–brand relationships with fundamental principles of human-to-human dialogue (Grice 1975; Jurafsky and Martin 2019; Searle 1969; Wiemann and Knapp 1975), and we introduce a "verbal embodiment" lens to better understand consumer–brand interactions mediated by conversational AI interfaces. This work sheds light on how altering the verbal embodiment of a brand through conversational interfaces impacts a consumer's experience with, and response to, the brand. It provides new insight on how enhanced verbal embodiment in AI-based conversational interfaces shape consumer–brand experiences in the marketplace (Brakus, Schmitt, and Zarantonello 2009; Holbrook and Hirschman 1982; Schmitt 2010).

In what follows, we first develop a theoretical framework that maps out the key properties of automated conversational interfaces, along with a set of hypotheses about the impact of these properties on consumers' interface perceptions, brand relationships, and brand-related behavior. Next, we present evidence from four studies that were designed to test this theorizing. We conclude with a discussion of how this work contributes to the emerging body of research on conversational AI in consumer–brand interactions, the theory-driven design of effective conversational interfaces in practice, and the broader implications for the design of consumer–AI interactions in the marketplace.

THEORY AND HYPOTHESES

Consumer–Brand Relationships as an Active Social Process

Language is "the mark of humanity and sentience" and dialogue is thought to represent "the most fundamental and specially privileged arena of language" (Jurafsky and Martin 2019). Dialogue is an active social process through which humans establish, define, and advance their relationships with others by creating shared meaning from experiences and a common understanding of the reality that surrounds them (Wood 1982). Seminal research on consumer–brand interactions has shown that consumers build relationships with brands similarly to how they build relationships with other humans (Fournier 1998; Muniz and O'Guinn 2001). These relationships develop through immediate interactions, and over time, shape consumers' subjective experience and interdependence with that brand with important consequences on consumer behavior (Brakus et al. 2009; Esch et al. 2006; Schmitt 2012).

With the proliferation of AI-based conversational interfaces, machines are entering a territory of natural language that was previously exclusive to humans. Prior work on "computers as social actors" (Nass, Steuer, and Tauber 1994) demonstrated that humans apply the same social norms to computers as they apply to other humans (Nass and Moon 2000). This suggests not only that the attribution of human likeness is a socially constructed process rather than a physical property (Lee and Nass 2003; Nass and Moon 2000; Zierau et al. 2022) but also that the design of technology-mediated consumer–brand interactions requires a better understanding of the fundamental principles governing these social interactions. We introduce the concept of *verbal embodiment* as a new theoretical lens to study how fundamental properties of human-to-human conversations can be harnessed to shape co-constructed and inherently social consumer–brand experiences through conversational Artificial Intelligence (AI).

Verbal Embodiment

Conversation is a collective activity in which two interacting parties must coordinate on both the process and content of a conversation (Clark and Brennan 1991; Grice 1989) to develop a relationship (Reis, Lemay, and Finkenauer 2017; Reis and Shaver 1988; Wiemann and Knapp 1975). In this research, we focus on and isolate essential properties that shape the *process* of a conversation, as opposed to other typical design features of the interface (e.g., a humanized avatar) or content of the conversation (e.g., emojis, typeface, formality of the language) that might influence attributions of embodiment (Li, Chan, and Kim 2019). Based on the concept of an "embodied system" as a "computing interface that mimics a human" (Cassell et al. 2000), we posit that a conversational interface becomes an embodied system by utilizing key principles of human dialogue structure. Such an embodied system operates not through visual embodiment (i.e., human face and body) but through verbal embodiment of human dialogue principles.

The social process of human dialogue has been conceptualized with respect to three core properties, highlighting that human dialogue requires that (1) the contributions happen in a reciprocal manner (turn-taking), (2) the interacting parties have control over their individual contributions (turn initiation), and (3) contributors acknowledge that they have received and understood the other party's contribution (grounding) (Jurafsky and Martin 2019). No matter what is being said during an interaction, these principles allow a conversation to happen in an organized and structured way (Holler et al. 2015). These principles are therefore independent of the content of the conversation and represent the process-related principles of a conversation that renders human dialogue an inherently social process (Clark and Brennan 1991; Holler et al. 2015; Levinson 2016; Op den Akker and Bruijnes 2012).

We focus on these three fundamental principles in the present research as they reflect core principles of the human dialogue *process* and are independent of the *content* of the conversation. The current research is therefore agnostic to the semantic meaning of individual contributions (i.e., speech acts) and only focuses on the interactive process used to establish rapport. Our work therefore aims at enhancing our understanding of how the *process* of human dialogue, through turn taking, turn initiation, and grounding, contributes to the social process of establishing a relationship with a brand.

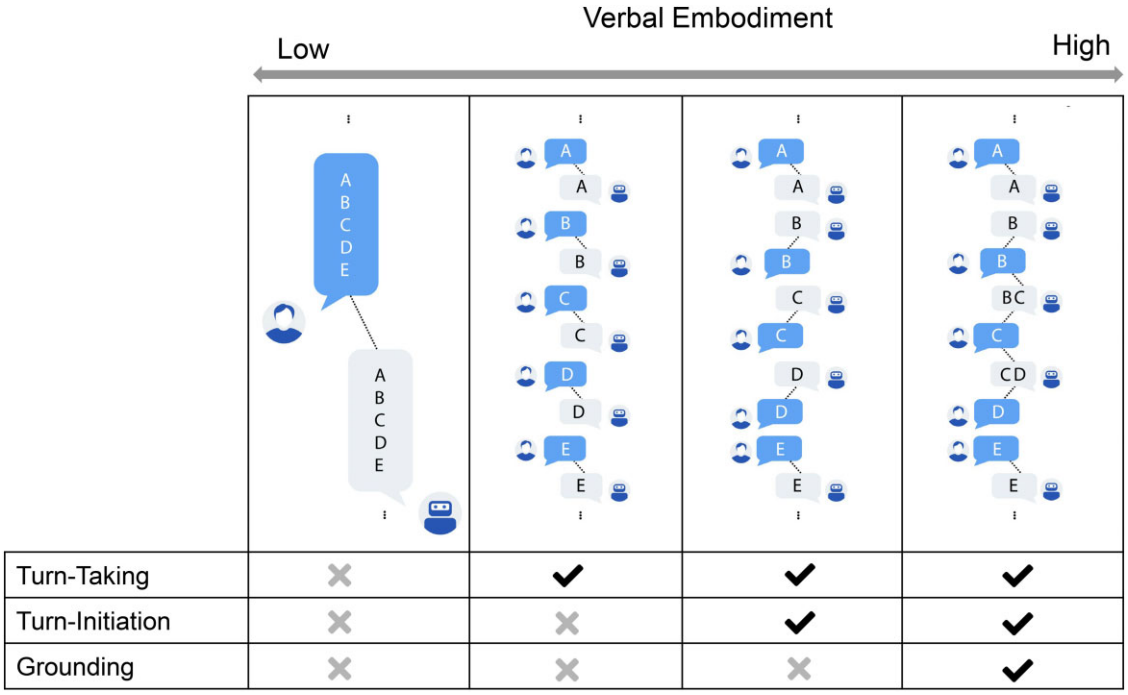
Figure 1 provides a schematic overview of how turn-taking, turn initiation, and grounding define the *conversational process* during a consumer–brand interaction. Comparing the first two panels shows that adding turn-taking to the interaction leads to a conversational flow with alternating contributions from the consumer and the conversational interface. In the third panel, the interface not only responds to the consumer, but it can also actively initiate subsequent turns by raising a new topic (symbolized by a different letter). Finally, in the fourth panel, the conversational interface not only responds to the consumer and initiates new turns, but it also grounds the consumer's contributions to signal understanding by restating or paraphrasing the consumer's previous input (symbolized by the letter representing the human's previous contribution). In what follows, we will elaborate on each of these three communication principles that make conversations social, co-constructed activities (Clark and Brennan 1991; Clark and Schaefer 1989; Searle 1969).

Turn-Taking. The reciprocal contribution to a dialogue, also called turn-taking, is a fundamental property of interpersonal communication. The knowledge of when to speak and what it means when someone interrupts or remains silent makes conversations a cooperative, social act (Grice 1975; Op den Akker and Bruijnes 2012). Turn-taking among humans is universal across languages, stemming from a combination of instinctive and learned behavior (Levinson 2016). Infants as young as 6 months try to mimic structural properties in a "conversation," even though they do not understand the spoken message content, while around the age of 9 months, turn-taking becomes more synchronized (Levinson 2016). Beyond the basic acts of recognizing and acting on the behavioral cues of one's conversational partner, turn-taking is a "truly responsive or co-constructed activity" (Cassell et al. 2001), reflecting a uniquely human ability.

Turn-taking in humans signals reciprocity and has important implications for defining the respective roles of conversational partners during an interaction (Wiemann and Knapp 1975), and for building a relationship between conversation partners (Reis and Shaver 1988). For

FIGURE 1

VERBAL EMBODIMENT THROUGH TURN-TAKING, TURN INITIATION, AND GROUNDING



NOTE.— The human and the bot icons symbolize the consumer and the AI-based conversational interface, respectively. Letters represent topics. Dots at the top and bottom indicate that this is merely a portion of the interaction—that is, there could be communication before and after what is illustrated here.

example, Sprecher et al. (2013) demonstrated that following a turn-taking protocol produces more favorable interpersonal outcomes compared to situations where conversational partners do not engage in turn-taking. Specifically, they showed that participants who disclosed reciprocally, as opposed to through a non-reciprocal monologue, reported greater liking, closeness, perceived similarity with the conversational partner, and enjoyment of the interaction.

Prior work on embodied systems revealed that human users tend to prefer interfaces that are capable of taking turns and that users experience greater comfort during an interaction where the interface is engaged in turn-taking (Cassell et al. 2001). Indeed, research on visual embodiment in information systems demonstrates that representing the reciprocal process of a dialogue through visual feedback (e.g., a nod or eye contact to give and take turns during a conversation) is more important than the actual visual display of emotions (e.g., smiling at the user) and has a greater impact on making the computing interface appear more human like and helpful (Cassell and Thórisson 1999). Against this background, we hypothesize that the capacity of an AI-based conversational interface to mimic human dialogue through a reciprocal process of turn-taking promotes consumers' attributions of human likeness to that interface.

> **H1:** Turn-taking between a consumer and a brand's AI-based conversational interface enhances the perceived humanness of the interface.

Turn Initiation. Turn initiation, also referred to as "conversational initiative" (Cassell et al. 2000; Jurafsky and Martin 2019; Walker and Whittaker 1995), represents the extent of control that each party has over a conversational interaction that involves turn-taking. For instance, if a conversational interface can only respond to a question posed by the user, then the interaction is fully controlled by that user. In human-to-human dialogue, it is more common for both parties to have equal control over the dialogue and thus be able to initiate turns, for example, by answering a question and then initiating a turn by asking a new question (Jurafsky and Martin 2019). This active contribution to a conversation is fundamental to the development of a closer relationship as it makes both parties responsive to the other's contributions during an

interaction (Reis and Shaver 1988). Such responsiveness is also central to the development of a closer bond between a parent and a child (Bowlby 1988).

In human–computer interactions, the ability to initiate turns can be thought of as reflecting the extent to which a system can control its own actions, which affects the perceived "socialness" of an interface (Nagao and Takeuchi 1994) and fosters a sense of cooperation during a conversation. Turn initiation is an important design feature of AI-based conversational interfaces and can range from low levels (e.g., a FAQ chatbot where the user fully controls the interaction and the interface only reacts to the input of the user) to high levels (e.g., a service chatbot that guides the interaction and autonomously responds to user input).

As human-to-human dialogue generally allows both parties to initiate turns during a conversation, we hypothesize that the capacity of a brand's AI-based conversational interface to initiate a turn helps endow the interface with enhanced levels of humanness.

> **H2:** Turn initiation by a brand's AI-based conversational interface enhances the perceived humanness of the interface.

Grounding. Grounding is the act of establishing a "common ground" by acknowledging that the listener has understood the speaker (Jurafsky and Martin 2019) through both verbal (e.g., "okay") and non-verbal (e.g., nodding or eye movement) acknowledgements (Richardson and Dale 2006). Such acknowledgements are made more explicit by repeating or paraphrasing what the speaker has said (McInnes and Attwater 2004). Thus, grounding does not add new content to a conversation but is a signal that the listener has not only received but also *understood* the speaker's previous contribution to the conversation. Jurafsky and Martin (2019) highlight that even simple objects provide nonlinguistic grounding signals, such as an elevator button that lights up once it has been pressed to acknowledge that the elevator has indeed been called. Yet, while these signals confirm that the object has received a command, linguistic grounding implies comprehension, which is an ability that remains difficult to confer to machines (Moritz et al. 2015).

By grounding each speaker's contributions, conversational turns become interconnected and contingent on what has been said previously (Sundar, Qian, and Bellur 2010). The act of grounding a message by acknowledging or paraphrasing confirms to the speaker that the other party is actively listening to what s/he is saying, suggesting that the other party is interested in, and paying attention to, what is being said, thereby fostering a sense of closeness and mutual understanding (Weger et al. 2014). These attributions can have important consequences for the interacting parties' perception of their relationship. For instance, Ramsey and Sohi (1997) found that grounding shapes a customer's perception of a salesperson and positively influences trust formation and consumers' inclination to interact with the same salesperson again in the future.

Grounding is also fundamental to the process of developing a relationship by acknowledging the other party's contribution to the conversation and thus establishing mutual knowledge (Clark and Brennan 1991; Reis et al. 2017). The need to feel understood features prominently in the literature on human relationships, as it helps to mitigate conflict (Weger 2005) and to form closer relationships by establishing shared thoughts and feelings (Reis and Shaver 1988). Taken together, we hypothesize:

> **H3:** Grounding between a brand's AI-based conversational interface and a consumer enhances the perceived humanness of the interface.

Leveraging Turn-Taking, Turn Initiation, and Grounding to Enhance Brand Intimacy

In human-to-human relationships, the formation of intimacy is a critical social process that serves as the foundation for a sense of mutual understanding between two entities (Clark and Schaefer 1989; Reis et al. 2017; Reis and Shaver 1988). The fact that greater intimacy in relationships is often characterized by more extensive self-disclosure (Sprecher and Hendrick 2004), reciprocity (McAdams, Jackson, and Kirshnit 1984), and positive affect (Lippert and Prager 2001), supports the idea that intimacy is closely associated with the formation and deepening of interpersonal connections. Reis and Shaver (1988) describe intimacy as a dynamic process that requires the active contribution of both parties to create mutual understanding, care, and validation. Thus, intimacy is an active process and includes a concern for the welfare of, high regard for, and intimate communication with, the other entity (Sternberg and Grajek 1984).

Building on the idea that consumer–brand relationships are based on similar social processes as interpersonal relationships (Fournier 1998), the development of intimacy with a brand is a dynamic process that requires interactive communication (Thorbjørnsen et al. 2002). Unlike other brand-related constructs, such as self-brand connection or brand attachment, where the focus is on the extent to which the brand is part of a consumer's identity (Escalas 2004; Park et al. 2010), brand intimacy refers to the extent of reciprocity and understanding between the consumer and the brand (Fournier 1998). Such perceived intimacy is an important antecedent of other brand-related outcomes such as brand love (Carroll and Ahuvia 2006; Rauschnabel and Ahuvia 2014) and brand attachment (Thomson, MacInnis, and Park 2005). The *brand intimacy* construct captures the degree to which consumers experience a (developing) bond with the brand and feel that the brand understands and cares about them (Thorbjørnsen et al. 2002). This sense of understanding as perceived by the consumer results in a

feeling of being cared about and validated by the brand (Natarelli and Plapler 2017; Thorbjørnsen et al. 2002).

Based on this prior work, we propose that enhanced verbal embodiment (via turn-taking, turn initiation, and grounding) is a key mechanism through which a brand's AI-based conversational interface can establish a closer connection with the consumer. Specifically, turn-taking in a conversation signals reciprocity and serves as the basic ingredient to shape a connection between conversational partners. Turn-taking without turn initiation implies a one-sided dialogue, whereas turn initiation further signals active participation and responsiveness by each party. The process of grounding entails making reference to the content of previous turns and thus enhances feelings of understanding and closeness by establishing inherent connections between individual turns. These verbal embodiments in a brand's conversational interface should facilitate the development of a deeper bond between the consumer and the brand. Thus, we hypothesize that interactions with an AI-based interface promote more intimate consumer–brand relationships, by enhancing perceived humanness due to the ability to take and initiate turns and to increase mutual understanding through grounding.

> **H4:** The perceived humanness of a brand's AI-based conversational interface has a positive effect on consumers' experienced brand intimacy.

The Impact of Enhanced Brand Intimacy on Behavioral Brand Outcomes

The semantic origin of the word "intimate" stems from the Latin word "intimare," meaning "to make the innermost known." Greater intimacy in human-to-human relationships is a key driver of interpersonal influence (Cialdini and Sagarin 1994; Finch and Cialdini 1989; Goldstein, Cialdini, and Griskevicius 2008; Maner et al. 2002). Intimacy in human relationships is also a critical antecedent of attachment to another entity (Baumeister and Leary 1995; Deci and Ryan 2000; La Guardia et al. 2000) and commitment to a relationship (Gilbert 1976).

Research in marketing has shown that intimacy with a brand strengthens the bond consumers feel toward it (Fournier 1998), with important consequences on brand-related behavior ranging from an increased commitment toward the brand to a greater willingness to pay to obtain it (Thomson et al. 2005). Thus, we hypothesize that the level of brand intimacy that consumers perceive as a result of interacting with a brand via its AI-based conversational interface has a positive impact on behavioral outcomes that are desirable from the brand's perspective. The brand-related behaviors that we examine in this research range from ones that are specific to the transaction (e.g., consumers' acceptance of a recommendation or their willingness to pay a price premium) to behaviors that go beyond the current transaction (e.g., consumers' enduring commitment to and advocacy of the brand).

> **H5:** Greater brand intimacy has a positive effect on consumers' (1) inclination to accept brand recommendations, (2) willingness to pay a price premium for the brand's products, (3) motivation to advocate the brand to others, and (4) brand loyalty.

The Moderating Role of Context-Specific Need for Mutual Understanding

Against the background of our conceptualization of brand intimacy as a co-created social construct, we propose that not all contexts are equally conducive to consumers developing such a sense of intimacy based on their interactions with a brand. In particular, the nature of the setting where a consumer–brand interaction occurs influences the extent to which a deep, nuanced understanding of the other party is necessary for the interaction to result in a mutually satisfying outcome. Interactions that entail a nuanced understanding of the consumer's unique preferences tend to require a higher level of mutual understanding (Castelo, Bos, and Lehmann 2019; Longoni et al. 2019) as consumers may be concerned about whether the brand adequately understands their idiosyncratic, subjective preferences. Contexts in which consumers reveal more subjective aspects of their "inner self" have the potential for greater intimacy than contexts in which consumers merely reveal objective, outwardly evident facts (Reis and Shaver 1988).

We theorize that this characteristic of the context in which consumer–brand interactions take place is a key moderator of the predicted effects. Specifically, we hypothesize that the conversational properties of a brand's AI-based conversational interface have less of an impact on the consumer's experience of brand intimacy in settings where there is a lower need for mutual understanding between the brand and the consumer.

> **H6:** The positive effects of the conversational properties of a brand's AI-based conversational interface on brand intimacy are attenuated in settings that do not require a high degree of mutual understanding.

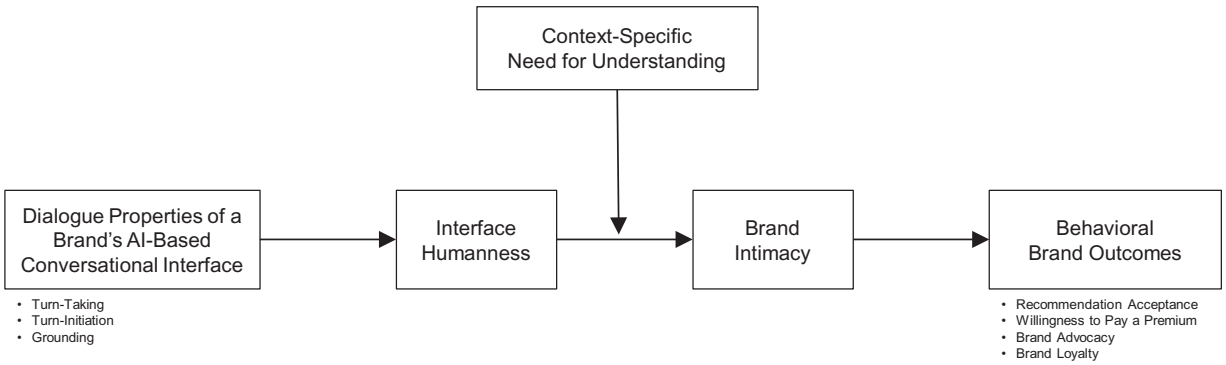
The conceptual model shown in figure 2 provides a concise, high-level summary of our theorizing by depicting the hypothesized relationships among the key constructs.

OVERVIEW OF STUDIES AND EXPERIMENTAL SETTING

We present evidence from four studies that were designed to test our theorizing by isolating the specific dialogue properties of an AI-based conversational interfaces and assessing how they shape consumer–brand relationships. In all studies, we employ tightly controlled but

FIGURE 2

CONCEPTUAL MODEL



realistic experimental paradigms in which participants engaged in real interactions with AI-based conversational interfaces. Details on the conversational setup and experimental paradigms can be found in our accompanying web appendix and the datasets of all studies can be accessed on the OSF data repository: https://osf.io/ejvhu/?view_only=181aae699eff42ae877054fe9e846ffc.

Our overarching testing paradigm is as follows: all studies anchor on the baseline effect of turn-taking and progressively test how it is augmented by the additive effects of turn initiation and grounding (see also figure 1). Studies 1 and 2 establish the baseline effect of turn-taking and turn initiation on attributions of humanness and subsequent effects on brand intimacy. Study 1 orthogonally manipulates turn-taking and turn initiation between subjects and demonstrates that enhanced turn-taking and turn initiation increase the perceived humanness of a brand's interface and that greater interface humanness, in turn, increases perceived brand intimacy (study 2). Studies 3 and 4 further assess the role of grounding and the moderating role of mutual understanding. Study 3 demonstrates that grounding further enhances the perceived humanness of a brand's interface and leads to greater perceived intimacy with the brand. Study 4 shows that the positive effects of the conversational properties of a brand's AI-based interface on brand intimacy are attenuated in settings that require lower levels of mutual understanding.

In all studies, we used custom-developed AI-based conversational interfaces that enabled us to manipulate the dialogue properties of turn-taking, turn initiation, and grounding with maximum experimental control. (See web appendix 2 for details and examples.)

STUDY 1

Study 1 examined the impact of the two key properties of a brand's AI-based conversational interface, turn-taking and turn initiation, on attributions of human likeness and perception of the brand. Based on our theorizing, we predict that both the ability to take turns during the interaction with the interface (hypothesis 1) and the interface's ability to autonomously initiate a turn (hypothesis 2) increase the perceived humanness of the interface and lead to more positive affective consumer reactions toward the brand.

Design and Procedure

A total of 226 consumers recruited via MTurk participated in this study (M_{Age}=30.89, SD_{Age}=15.00; 59% females). They were randomly assigned to a 2 (presence vs. absence of turn-taking) × 2 (presence vs. absence of turn initiation) between-subjects experimental design. Participants were invited to take part in a study on consumer design preferences and informed that they would be interacting with an automated conversational interface. We created a fictitious design brand called "AI.Design" and participants were led to believe that the purpose of the study was to better understand consumers' design preferences for everyday objects. Next, participants completed a simple design task in which they chose between one of three feature levels for each of eight design features (e.g., geometric shape, fill pattern, line pattern, and ornaments; see web appendix 3.2 for details).

In the conditions with turn-taking, the interface first provided participants all three feature levels and then asked them to indicate their preferred feature level by typing their answer into the chat console. After participants provided their answer, the system went on with the next feature and continued sequentially until all feature levels were specified. In the conditions without turn-taking, the interface provided identical features and feature levels in exactly the same order, but all options were presented simultaneously. After the interface presented all features and feature levels,

participants were asked to specify their preferred feature level by typing their answer into the chat console.

We implemented the turn-initiation manipulation as follows: In the turn-initiation-present condition, the interface directly asked the next question in the sequence of feature-level evaluations, thus playing an active role in the interaction. By contrast, in the turn-initiation-absent condition, the interface had a passive role and participants controlled the flow of the interaction by clicking a "Next" button to see the next feature and corresponding feature levels. Note that this manipulation did not affect participants' input to the design question, and they were not able to select a different option once they had made a selection and before clicking the "Next" button.

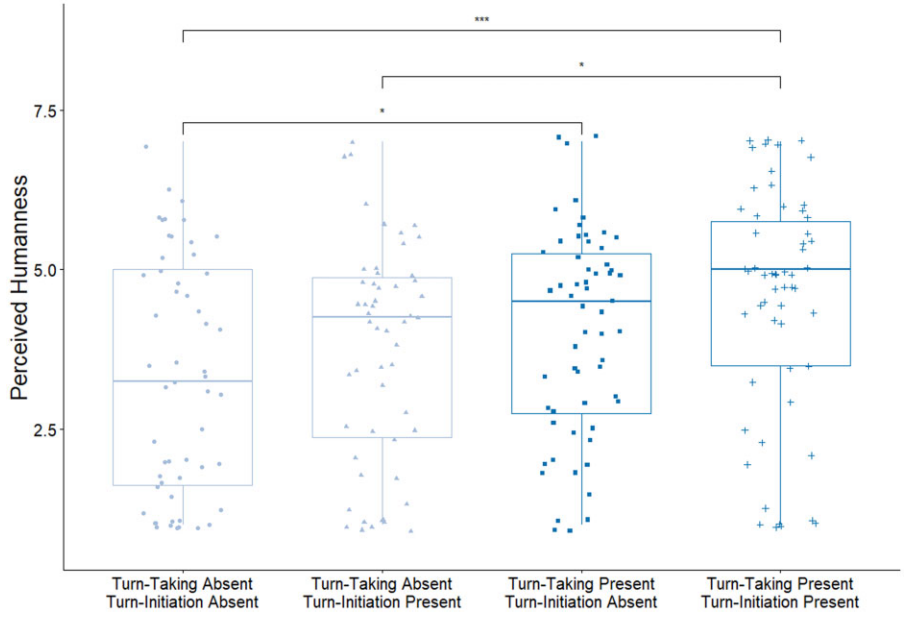
Measures. Immediately after participants completed the design task, we assessed the perceived humanness of the interface (Rijsdijk, Hultink, and Diamantopoulos 2007) using four items: "This system . . . has human properties, . . . is like a person, . . . behaves like a human being, . . . has its own character;" 1 = "not agree at all" to 7 = "fully agree;" $\alpha = 0.96$. Next, we used a qualitative, open-response technique to assess consumers' spontaneous brand-related thoughts. Participants first wrote down their spontaneous thoughts or feelings related to the brand and then evaluated the valence of each of these thoughts as negative (coded as −1), neutral (coded as 0), or positive (coded as 1). In the following analyses, we concatenated all text entries of the thought-listing task into a single text vector. We used the AFINN lexicon (Nielsen 2011) for all subsequent sentiment analyses to evaluate consumers' brand sentiment conditional on our experimental manipulations.

Results

In support of our theorizing, we find that both turn-taking (hypothesis 1) and the interface's ability to initiate turns in a conversation (hypothesis 2) significantly influenced how participants perceived the interface. As illustrated in figure 3, turn-taking significantly increased perceived humanness ($M_{\text{TurnTaking_Present}} = 4.28$, $M_{\text{TurnTaking_Absent}} = 3.52$; $F(2, 222) = 10.561$, $p < .01$, η^2=0.05). Furthermore, turn initiation by the system further increased the level of perceived humanness ($M_{\text{TurnInitiation_Present}} = 4.15$, $M_{\text{TurnInitiation_Absent}} = 3.68$; $F(2, 222) = 4.137$, $p < .05$, η^2=0.02). The interaction between the two factors was not significant ($F(1, 222) = 0.06$, $p > .80$), indicating that both properties act as independent forces to increase consumers' perception of humanness with the effect size of turn-taking being moderately stronger ($d = 0.43$) than that of turn initiation ($d =$

FIGURE 3

TURN-TAKING AND TURN INITIATION AS DRIVERS OF HUMANNESS PERCEPTIONS



NOTE.— ***$p < .001$, **$p < .01$, *$p < .05$.

0.26). These findings suggest that both turn-taking and turn initiation contribute independently to the perception of the humanness of the interface.

Next, we examined participants' spontaneous thoughts about the "AI.Design" brand. In line with our theorizing, and mirroring the effects on perceived humanness, both turn-taking ($M_{\text{TurnTaking_Present}} = 2.64$, $M_{\text{TurnTaking_Absent}} = 0.53$; $F(2, 222) = 15.522$, $p < .001$, η^2=0.06) and turn initiation ($M_{\text{TurnInitiation_Present}} = 2.47$, $M_{\text{TurnInitiation_Absent}} = 0.76$; $F(2, 222) = 10.202$, $p < .01$, η^2=0.04) led to significantly more positive brand sentiment, while the interaction between the two factors was not significant ($F(2, 222) = 0.212$, $p = .64$).

To further illustrate the pattern of results based on consumers' expressed brand sentiment, figure 4 shows the relationship between perceived humanness and brand sentiment across each experimental condition. Each observation is color coded based on the corresponding sentiment score (using the AFINN lexicon, see Measures) and the most frequent sentiment for each data point (i.e., participant). These findings reveal that greater perceived humanness is associated with a more positive brand sentiment and that the greatest density of positively valenced brand sentiment occurs when both turn-taking and turn initiation are present.

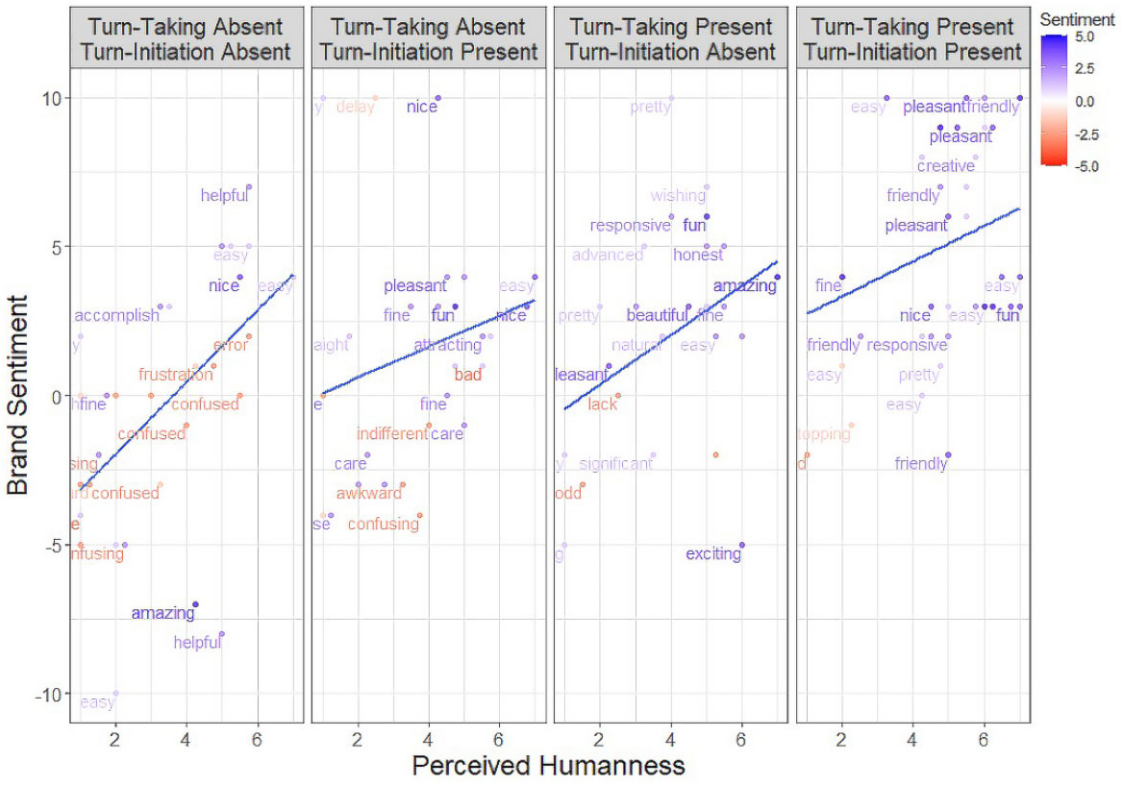
Mediation Analysis. We estimated a mediation model with 5,000 bootstrap samples contrasting the interface with both turn-taking and turn initiation versus all other conditions. Perceived interface humanness served as the mediator and brand sentiment as the dependent variable. This analysis revealed a positive, significant indirect effect (β_{Indirect}=0.90, 95% CI: [0.31; 1.55]), indicating that the conversational properties of a brand's AI-based interface enhance brand sentiment via an increase in the perceived humanness of the interface.

Discussion

The findings of this study provide initial evidence for our theorizing about how turn-taking and turn initiation promote greater perceptions of interface humanness and more positive affective consumer reactions toward the brand.

FIGURE 4

BRAND SENTIMENT AS A FUNCTION OF PERCEIVED HUMANNESS



STUDY 2

Study 2 was designed with three key objectives in mind. First, while study 1 used a qualitative measure to assess consumers' relationship with and positive affect toward the brand, study 2 provides a more direct test of our hypothesis that turn-taking enhances consumers' experienced brand intimacy (hypothesis 4). Second, this study probes whether the properties of conversational interfaces can have an impact on product choices, and it does so by examining consumers' acceptance of a product recommendation they receive during their interaction with a brand's conversational interface. Third, we investigate potential alternative accounts of the effect of turn-taking on brand intimacy based on perceived novelty, ease of use, or task engagement.

Design and Procedure

All methods and statistical analyses for this study were pre-registered (https://aspredicted.org/blind.php?x=jf9ti7). A total of 391 consumers (M_{Age}=31.17, SD_{Age}=11.51; 64% females), recruited through Prolific Academic, completed the study. They were randomly assigned to a turn-taking-present or turn-taking-absent conversational interface condition (see web appendix 2.2 for screen shots). Each participant was led through a shirt customization process for a fictional shirt brand called "SwissThread," answering questions on their body type, preference for shirt fit, hair and eye color, along with the customization of their preferred shirt type, color, and style before receiving a recommended shirt option (see web appendix 3.1 for details). All questions were identical and presented in the same order in both interface conditions. In the condition with turn-taking, participants sequentially specified their answer by taking one turn after the other, whereas those in the condition without turn-taking were asked all five profiling questions at once, before indicating their preferred options. In both conditions, the interface initiated turns. Participants were informed that they were interacting with an automated conversational interface.

After completion of the profiling task, the interface recommended a shirt in a specific style and color, which the participant could either accept or reject and choose a different style from a set of available options (web appendix 3.1). Participants then rated the perceived humanness of the interface using the same four items as in study 1 (α_{human} = 0.93, web appendix 5.1) and indicated how intimately they felt connected to the brand using three items (Thorbjørnsen et al. 2002: "I feel like SwissThread really cares about me," "SwissThread really listens to what I have to say," "I feel as though SwissThread really understands me," 1 = "I do not agree at all" to 7 = "I fully agree"; $\alpha_{BrandIntimacy}$ = 0.90). Finally, to rule out that the effects can be explained by a mere novelty effect or that the interface is easier to use or evokes greater task involvement, we measured the perceived novelty when using the interface (Guiry, Mägi, and Lutz 2006; sample item: "The system offers novel experiences," $\alpha_{Novelty}$ = 0.88), task involvement (Escalas, Moore, and Edell 2004; sample item: "The system really intrigued me," $\alpha_{Involvement}$ = 0.90), and ease of use (Keller and Block 1997; sample item [reverse coded]: "The system required a lot of effort," α_{Ease} = 0.86). Details on all scale items can be found in web appendices 5.5–5.7.

Manipulation Check

At the end of the study, we assessed the effectiveness of the manipulation by asking participants to rate the perceived extent of turn-taking with the interface (adapted from Song and Zinkhan 2008; sample item: "The system facilitated a two-way conversation," $\alpha_{TurnTaking}$ = 0.90, web appendix 5.3). The interface manipulation had a significant effect on the perception of turn-taking ($M_{TurnTaking_Present}$ = 4.44, $M_{TurnTaking_Absent}$ = 3.82; $F(1, 389)$ = 16.87, $p < .001$, η^2=0.04), confirming that the manipulation was effective.

Results

Interface Humanness. In line with our prediction (hypothesis 1) and replicating the results from study 1, participants perceived a conversational interface with turn-taking as significantly more human like compared to the interface without turn-taking ($M_{TurnTaking_Present}$ = 4.59, $M_{TurnTaking_Absent}$ = 3.73; $F(1, 389)$ = 30.62, $p < .001$, η^2=0.07).

Brand Intimacy. Furthermore, we also find that consumers experienced a significantly more intimate relationship with the brand in the turn-taking condition compared to the no-turn-taking condition ($M_{TurnTaking_Present}$ = 4.52, $M_{TurnTaking_Absent}$ = 3.94; $F(1, 389)$ = 13.62, $p<.001$, η^2=0.03). In line with our theorizing (hypothesis 4), a mediation model with 5,000 bootstrap samples confirmed that the effect on greater perceptions of brand intimacy was significantly mediated by the greater perception of humanness of the interface ($\beta_{Indirect}$=0.66, 95% CI: [0.42; 0.90]), rendering the residual direct effect non-significant (p = .49) and thus indicating full mediation.

Recommendation Acceptance. Mirroring the findings for brand intimacy and perceptions of humanness, participants in the turn-taking-present (vs. -absent) condition were significantly more likely to accept the recommended shirt at the end of the shopping task ($P_{TurnTaking_Present}$ = 57.6%, $P_{TurnTaking_Absent}$ = 37.2%, z = 4.01, $p < .001$).

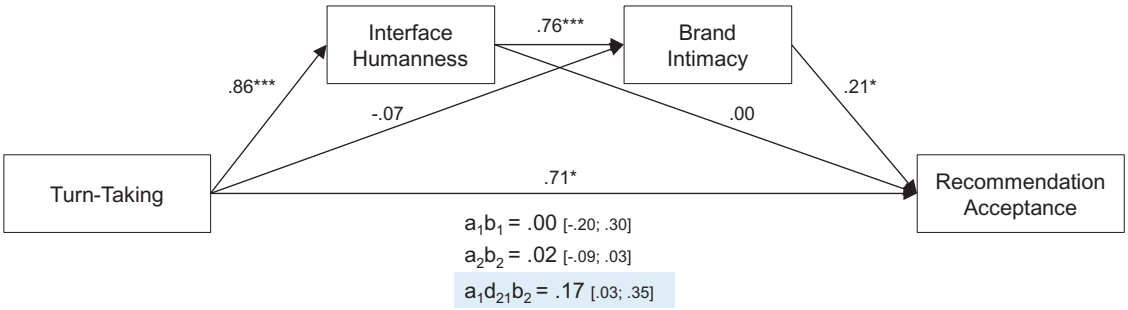
Sequential Mediation. Next, we estimated a sequential mediation model (5,000 bootstrap samples). The interface manipulation served as the independent variable, perceived

humanness and brand intimacy as proximal and distal sequential mediators, and acceptance of the recommended product as the dependent variable. The results (figure 5) show that, in support of our theorizing, the positive effect of turn-taking on recommendation acceptance was sequentially mediated by greater perceived interface humanness and increased brand intimacy ($\beta_{Indirect}$=0.17, 95% CI: [0.03; 0.35]).

To further illustrate the pattern of results, the density plots shown in figure 6 demonstrate that the probability mass of recommendation acceptance is systematically shifted toward higher levels of perceived interface

FIGURE 5

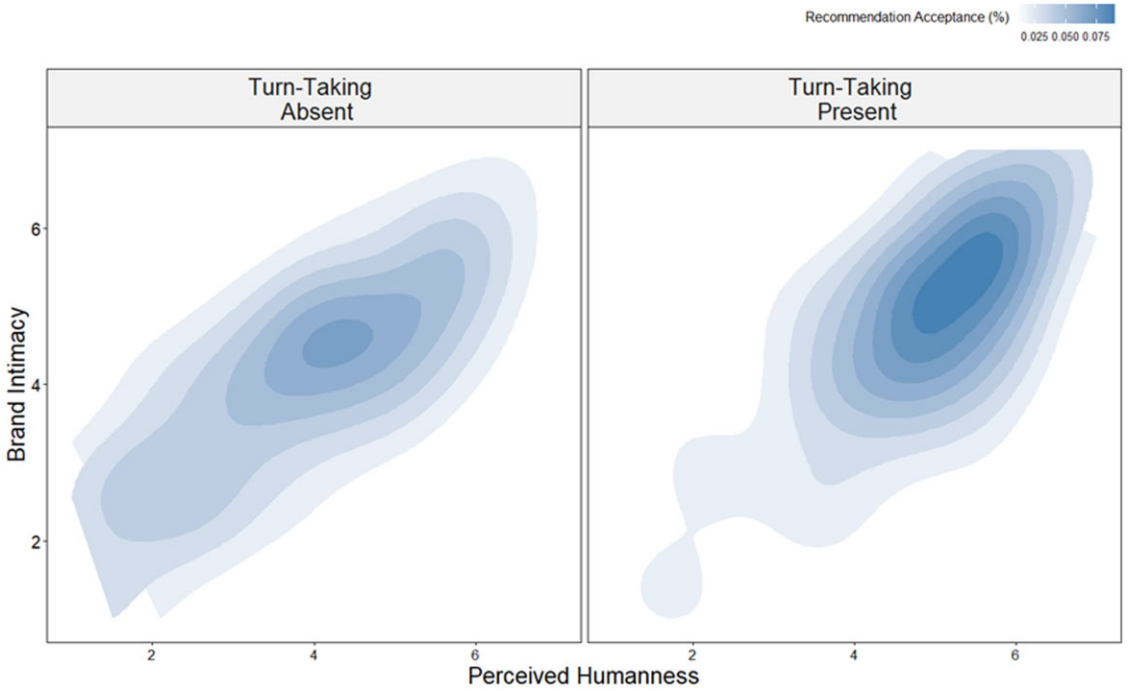
SEQUENTIAL MEDIATION MODEL



NOTE.— ***$p < .001$, **$p < .01$, *$p < .05$.

FIGURE 6

RECOMMENDATION ACCEPTANCE AS A FUNCTION OF PERCEIVED HUMANNESS AND BRAND INTIMACY



humanness and brand intimacy in the turn-taking-present compared to the turn-taking-absent conditions.

Robustness Tests. We performed a series of robustness tests to further assess whether the effects of turn-taking on brand intimacy are explained by a mere novelty effect, differences in ease of use perceptions, or task engagement. We found that the turn-taking manipulation had a significant impact on novelty ($M_{\text{TurnTaking_Present}} = 4.76$, $M_{\text{TurnTaking_Absent}} = 4.38$; $F(1, 389) = 6.11$, $p < .05$, $\eta^2 = 0.02$), ease of use ($M_{\text{TurnTaking_Present}} = 5.03$, $M_{\text{TurnTaking_Absent}} = 4.34$; $F(1, 389) = 26.68$, $p < .001$, $\eta^2 = 0.06$), and task involvement ($M_{\text{TurnTaking_Present}} = 4.16$, $M_{\text{TurnTaking_Absent}} = 3.78$; $F(1, 389) = 4.87$, $p < .05$, $\eta^2 = 0.02$). However, supporting the robustness of our theorizing, the effect of turn-taking on perceived interface humanness remained significant even after controlling for these three variables ($t(386) = 4.617$, $p < .001$). The same was true for the effect of interface humanness on brand intimacy ($t(386) = 14.624$, $p < .001$). We also assessed the robustness of the humanness mechanism using a series of Davidson–MacKinnon tests for non-nested models (Davidson and MacKinnon 1981). If ease of use, novelty, and involvement were sufficient to explain the increase in brand intimacy, then including the fitted values of the model testing humanness perceptions should provide no significant improvement in explanatory power. However, if adding the fitted values of humanness perception did improve prediction accuracy, we can conclude that the alternative mechanism model does not contain the correct set of regressors to predict brand intimacy (see also Greene 1993). While including the fitted values of ease of use, involvement, and novelty in the perceived humanness model revealed a significant improvement in model fit ($\theta_{\text{HumannessModel}}$ + Fitted(AlternativeModel), $t = 8.345$, $p < .001$), we found a comparatively stronger, significant improvement when accounting for the fitted values of the humanness mechanism in the alternative mechanism model ($\theta_{\text{AlternativeModel}}$ + Fitted(HumannessModel), $t = 14.624$, $p < .001$). This result highlights the importance of perceived interface humanness in understanding how the properties of an AI-based conversational interface affect brand intimacy. Additional robustness tests leveraging a parallel mediation model with these alternative mechanisms further corroborate our results (web appendix 6.2). Finally, we also controlled for the possibility that different consumer groups may be affected differentially by the two interface conditions. In short, we found that all of the reported effects remained significant and none were moderated by differences in participants' age (all p values $>.64$) or gender (all p values $>.13$), suggesting that the effect is robust across a broad range of consumer demographics.

Discussion

The results of study 2 demonstrate the pivotal role of turn-taking in the ability of a brand's AI-based conversational interface to promote perceptions of humanness, enhance brand intimacy, and influence consumer choice. The findings also rule out a series of potential alternative mechanisms via novelty attributions, task involvement, or ease of use. However, as we observed a significant increase across each of these process accounts (increase in novelty, involvement, and ease of use), we acknowledge that the effects of conversational AI interfaces on consumers are multiply determined, which is also reflected in the partial mediation on downstream recommendation acceptance.

STUDY 3

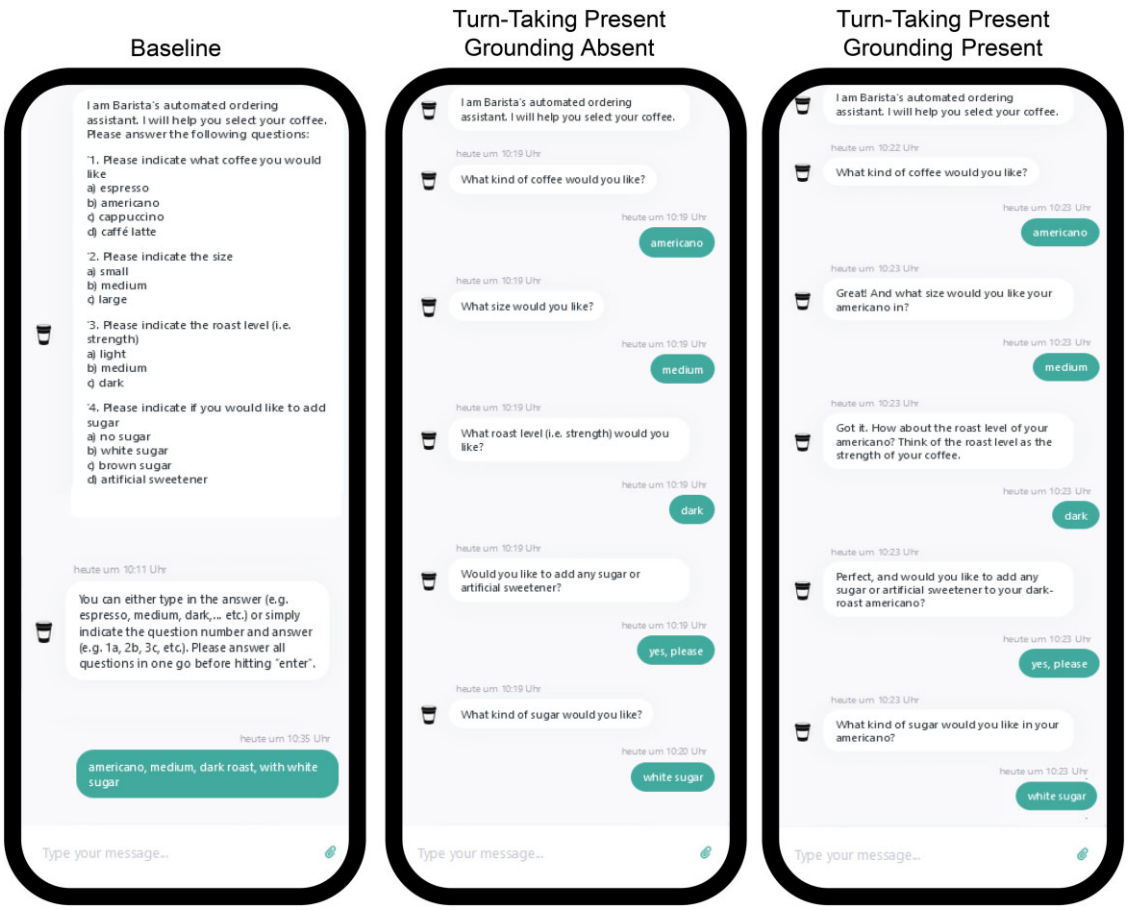
A hallmark of human communication and effective conversation is the signaling of mutual understanding through conversational grounding. One objective of study 3 was to examine whether semantic grounding affects perceptions of humanness and brand intimacy above and beyond turn-taking. In addition, this study was designed to provide further insight into the downstream consequences of interacting with a brand's AI-based conversational interface on brand-related behaviors, especially those related to future commitment to the brand (such as brand loyalty or consumers' willingness to advocate the brand to others).

Design and Procedure

We pre-registered all methods and statistical analyses for this study (https://aspredicted.org/SFJ_XSB). A total of 580 participants (M_{Age}=29.39, SD_{Age}=9.91; 72% females), recruited through Prolific Academic, completed the study. As illustrated in figure 7, they were randomly assigned to one of three interface conditions—baseline, turn-taking only, or turn-taking plus grounding. (An orthogonal manipulation of turn-taking and grounding in a full-factorial design is not possible since turn-taking is a prerequisite for grounding; Jurafsky and Martin 2019, as well as figure 1.) In all three conditions, participants were explicitly told that they are interacting with an automated conversational interface. Participants used a conversational interface to order a to-go coffee with the brand Barista. They were informed that they would be interacting with an automated conversational interface. This interface did not include any other humanizing features (e.g., a human-like avatar). Participants' task was to order their preferred to-go coffee. They did so by answering a total of 10 questions to specify their order in terms of 10 different features (such as size, roast, type of milk, and amount of sugar; see web appendix 3.3 for the full conversational flow). In all conditions, participants went through the same questions, in the same exact order, to place their coffee order. Upon

FIGURE 7

BARISTA BOT CONDITIONS



completion of the order process, the interface asked participants whether they would be willing to sign up for a loyalty program with Barista as a behavioral brand-related outcome measure.

As illustrated in figure 7, in the baseline condition, the interface presented all 10 customization questions at the same time and asked the customer to specify their coffee order by answering all questions in a single turn. In the turn-taking without grounding condition, the interface presented one question at a time, waited for the customer to respond, and then proceeded to the next question. Building on prior work by Jurafsky and Martin (2019), we manipulated conversational grounding as follows. First, we used trivial acknowledgments signaling understanding after receiving a user input (such as "noted" or "OK"). Second, we used paraphrasing and repeating back the user input to further strengthen the signal of the interface that it has received and recorded the user input. For example, if the user answered "espresso" to the interface's question "What kind of coffee would you like?" the interface would record the user input and integrate it into the following question by saying "And what size would you like your *espresso* in?." Finally, halfway through the conversational flow—as well as at the end of the ordering process—the interface summarized and repeated back the desired order to further strengthen the signal that the interface had correctly understood the customer's order. These semantic properties did not alter turn-taking as they did not warrant a response from the user. Otherwise, the content and sequence of questions were identical across conditions.

As in study 2, we measured the perceived humanness of the interface ($\alpha_{human} = 0.93$) and brand intimacy ($\alpha_{BrandIntimacy} = 0.85$) using the same items. Next, we measured consumers' willingness to pay a price premium using four items (Jones, Taylor, and Bansal 2008; sample item: "I am likely to pay a little more for Barista"; $\alpha_{PayPremium} = 0.80$) and willingness to advocate the brand to others using three items (Maxham and Netemeyer 2003;

sample item: "I would recommend Barista as a coffee brand to my friends"; $\alpha_{RecommendIntent} = 0.96$). Finally, we assessed participants' loyalty intention based on three items (adapted from Han, Kwortnik, and Wang 2008; sample item: "I'd like to continue using Barista in the future"; $\alpha_{PurchaseIntent} = 0.94$). In addition to our primary dependent variables, we also measured consumers' brand rating using a single response item ("We will post a review of the customer experience people had with this customer service on a separate, crowd-sourced customer review platform. Please provide a star rating of the services below"; 5-point star rating scale). (Details on all of the scales can be found in web appendices 5.8–5.11.)

Manipulation Check

To assess the effectiveness of the experimental manipulation, we measured participants' perceived degree of grounding based on five items (adapted from Pierre and David 2006; sample item: "The system provided feedback of having perceived my input," $\alpha_{Grounding} = 0.89$, web appendix 5.4), as well as their perceived degree of turn-taking based on the same items as in the previous study 2 ($\alpha_{TurnTaking} = 0.91$). Supporting the effectiveness of our manipulation, the interface condition had a significant effect on the perception of grounding ($F(2, 577) = 95.68$, $p < .001$, η^2=0.25). Follow-up contrasts with Tukey HSD correction confirmed that the interface condition with grounding was perceived as significantly greater in grounding compared to both the turn-taking without grounding condition and relative to baseline ($M_{TurnTaking_GroundingPresent} = 5.99$, $M_{Baseline} = 4.36$; $t = 12.005$, $p < .001$; $M_{TurnTaking_GroundingAbsent} = 4.42$; $t = 11.924$, $p < .001$), while there was no difference in perceived grounding between turn-taking without grounding and baseline conditions ($p = .912$), supporting the effectiveness of the experimental manipulation. In addition, and in line with study 2, the interface condition also had a significant effect on the perception of turn-taking ($F(2, 577) = 12.05$, $p < .001$, η^2=0.04). Specifically, and supporting the effectiveness of our manipulation, follow-up contrasts with Tukey HSD correction confirmed that the two interface conditions with turn-taking were perceived as significantly greater in turn-taking relative to baseline ($M_{TurnTaking_GroundingPresent} = 4.49$, $M_{Baseline} = 3.61$; $t = 4.839$, $p < .001$; $M_{TurnTaking_GroundingAbsent} = 4.19$; $t = 3.234$, $p < .01$), while there was no difference in perceived turn-taking between the turn-taking with and without grounding ($p = .213$).

Results

Interface Humanness. A one-way ANOVA with perceptions of humanness as the dependent variable revealed a significant effect of interface type ($F(2, 577 = 55.420$, $p < .001$, η^2=0.16; figure 8). In line with our theorizing (hypothesis 3), follow-up contrasts with Tukey HSD correction comparing the turn-taking with versus without grounding conditions revealed a significant positive effect of grounding on the perceived humanness of the interface ($M_{TurnTaking_GroundingPresent} = 4.74$, $M_{TurnTaking_GroundingAbsent} = 3.78$, $t = 6.169$, $p < .001$). Moreover, each of the two turn-taking conditions yielded greater perceived interface humanness than the baseline condition ($M_{Baseline} = 3.07$; both p values $< .001$).

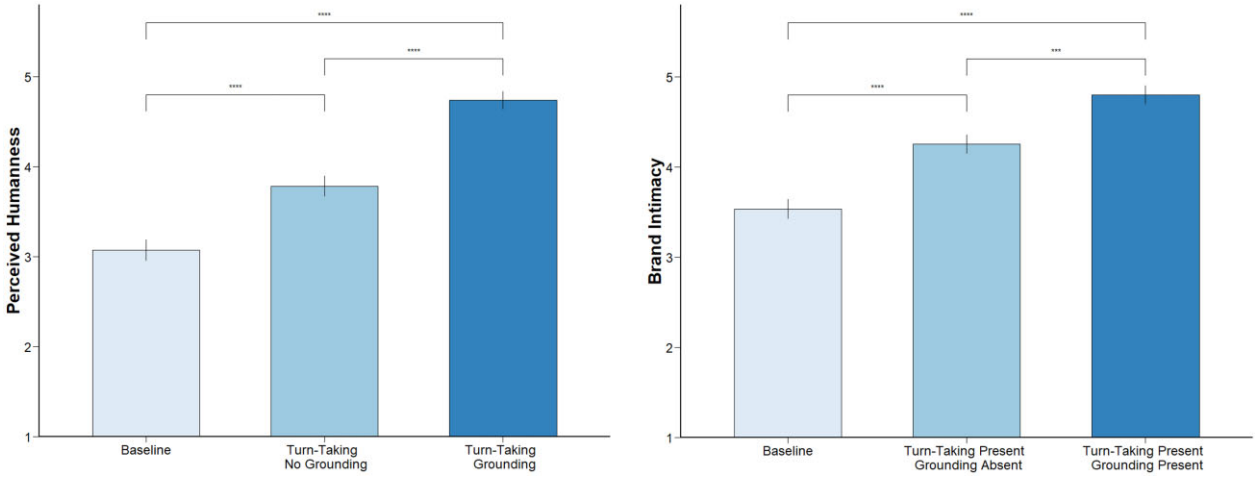
Brand Intimacy. The type of interface also had a significant overall effect on perceptions of brand intimacy ($F(2, 577) = 34.694$, $p < .001$, η^2=0.11). In line with our theorizing, follow-up contrasts with Tukey HSD correction revealed that brand intimacy was significantly higher in the turn-taking with grounding condition when compared to both, the turn-taking without grounding condition ($M_{TurnTaking_GroundingPresent} = 4.80$, $M_{TurnTaking_GroundingAbsent} = 4.25$, $t = 3.704$, $p < .001$) and to baseline condition ($M_{Baseline} = 3.53$, $t = 8.316$, $p < .001$). Corroborating the findings of study 2, brand intimacy was also significantly higher in the turn-taking without grounding condition compared to the baseline condition ($t = 4.758$, $p < .001$). These findings demonstrate that grounding not only enhances perceived interface humanness but also further promotes feelings of brand intimacy (above and beyond the effect of turn-taking).

Orthogonal Mediation Contrasts. Next, we estimated a mediation model using orthogonal contrasts (Helmert contrasts; 5,000 bootstrap samples). The interface condition served as the independent variable, brand intimacy as the dependent variable, and perceptions of humanness as the mediating process variable. Corroborating our results from study 2, we find that the positive effect of turn-taking (with and without grounding) compared to the baseline interface on brand intimacy was significantly mediated via first enhancing perceptions of humanness ($\beta = 1.19$, $t = 8.600$, $p < .001$) which subsequently led to a significant increase in brand intimacy ($\beta = 0.61$, $t = 20.052$, $p < .001$), leading to a positive indirect effect with a confidence interval excluding zero ($\beta_{Indirect}$=0.73, 95% CI: [0.55; 0.92]). Similarly, further testing the orthogonal contrast by comparing the interface employing turn-taking with versus without grounding led to the previously reported increase in brand intimacy (β=0.55, $t = 3.704$, $p < .001$), but this effect was significantly mediated via first enhancing perceptions of humanness (β =0.96, $t = 6.169$, $p < .001$) with a positive indirect effect through perceived humanness ($\beta_{Indirect}$=0.59, 95% CI: [0.39; 0.79]). A figure displaying this mediation model can be found in web appendix 6.3.

Path Model. We estimated a path model to further assess the impact of brand intimacy on behavioral brand outcomes (hypothesis 5). Specifically, we used the lavaan

FIGURE 8

IMPACT OF TURN-TAKING AND GROUNDING ON PERCEIVED INTERFACE HUMANNESS AND BRAND INTIMACY



NOTE.— ****$p < .0001$, ***$p < .001$, **$p < .01$, *$p < .05$.

package in R with robust standard errors to model the effect of grounding (turn-taking only coded as −1, baseline coded as 0, and turn-taking plus grounding coded as 1) on humanness; the influence of greater levels of humanness on brand intimacy; and the role of brand intimacy in driving subsequent brand-related behaviors. As summarized in figure 9, and mirroring the main effect findings, a greater extent of grounding enhanced perceived humanness (β=0.941, z=5.994, p<.001), which in turn increased consumers' experienced brand intimacy (β=0.640, z=20.798, p<.001) and switched off the direct effect of grounding (p = .565). Most importantly, brand intimacy had a significant positive effect on consumers' willingness to pay a price premium (β =0.501, z=14.960, p<.001), future purchase intentions (β=0.623, z=16.195, p<.001), and brand advocacy (β=0.682, z=18.913, p<.001). We also found that brand intimacy led to a significant boost in consumers' brand ratings (β=0.153, z=4.023, p<.001). These model results were robust, even after controlling for differences in participants' age and gender, as indicated by higher information criteria compared to our main model ($BIC_{MainModel}$: 9,300.3; $BIC_{ControlModel}$: 9,306.0).

Discussion

The results of study 3 demonstrate that grounding leads to greater perceived humanness of a conversational interface and significantly enhances the experience of brand intimacy. Furthermore, study 3 also demonstrates the strong subsequent impact of brand intimacy on a broad range of behavioral brand outcomes.

STUDY 4

The primary objective of study 4 was to test the moderating role of consumers' task-specific need for mutual understanding. Specifically, the study examines whether the positive impact of grounding in a brand's AI-based conversational interface on consumers' perceptions of brand intimacy is attenuated in settings that require lower levels of mutual understanding (hypothesis 6). In addition, this study demonstrates the robustness of the effects by ruling out potential demand effects that might have resulted from measuring perceived interface humanness before brand intimacy.

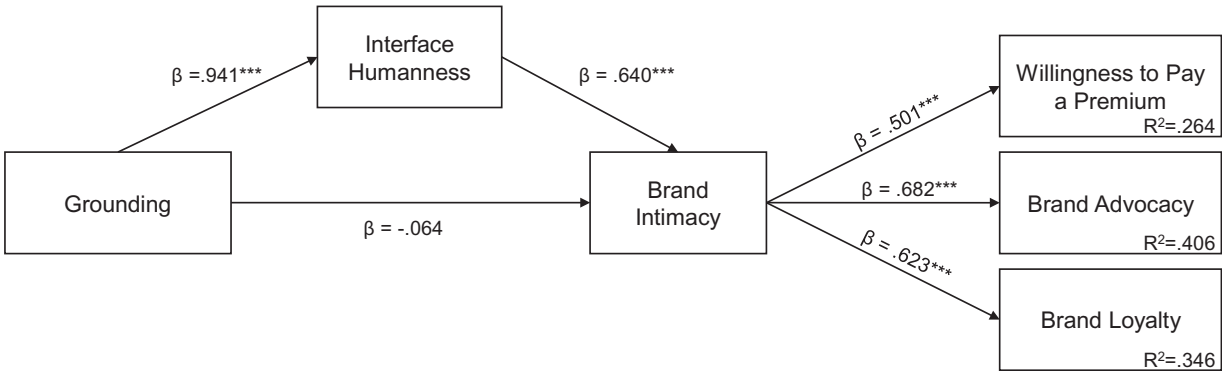
Design and Procedure

We pre-registered all methods and statistical analyses for this study (https://aspredicted.org/blind.php?x=G2J_BX8). A total of 574 participants (M_{Age}=39.66, SD_{Age}=12.57; 60% females), recruited from Prolific Academic, completed the study. They were randomly assigned to a 2 (properties of the interface: turn-taking without grounding vs. turn-taking with grounding) × 2 (need for understanding: low vs. high) between-subjects design.

According to our theorizing, the effect of the conversational properties of a brand's interface on consumers' perceptions of brand intimacy is attenuated in settings characterized by a low need for mutual understanding (hypothesis 6). This need should be lower in tasks that are more objective and require the provision of predominantly factual information by the consumer than in tasks that are

FIGURE 9

PATH MODEL



NOTE.— ***$p < .001$, **$p < .01$, *$p < .05$.

more subjective and require a more nuanced understanding of consumers' idiosyncratic preferences. As grounding increases the perceived humanness of an interface, and the perception that the brand is listening to and understanding consumer needs, its effect should be reduced in contexts where the nature of the dialogue is more objective versus subjective. Thus, we manipulated the need for mutual understanding via the objectivity of the task, building on Castelo et al. (2019). Specifically, we selected a task that was seen as relatively neutral from Castelo et al.'s (2019) objectivity ranking as our context—hiring an employee. Participants were introduced to an automated recruitment advisor from a fictitious candidate recruiting brand AgileRecruit. Building on Castelo et al. (2019), we manipulated task objectivity by describing the task of recruiting an employee either based on objective, quantifiable information or more subjective, intuitive pieces of information. Specifically, in the objective task condition, participants were told that "to find someone suitable, the recruitment agent needs to gather objective facts about the candidate's desired profile and hard skills," while in the subjective task condition, participants were told that "to find someone suitable, the recruitment agent needs to deeply understand the candidate's desired personality and soft skills." Participants were led through 13 profiling questions in which they were asked questions about prospective candidates. In the objective condition, these questions focused on demographic data and quantifiable information such as the number of years of work experience or the level of education. In the subjective condition, the 13 profiling questions focused on subjective, non-quantifiable data such as desired personality traits and working styles.

The following aspects were identical across conditions. Participants used a conversational interface to complete the respective profiling tasks (see web appendix 2.4 for sample stimuli). They were informed that they would be interacting with an automated conversational interface. This interface did not include any other humanizing features (e.g., a human-like avatar). Participants answered a total of 13 profiling questions and each profiling question contained the same number of choice levels in all conditions to control for potential differences in task complexity (web appendix 3.4).

Participants completed the profiling task using either an interface with only turn-taking but no grounding or one that included both turn-taking and grounding. In the turn-taking only condition, the interface presented one question at a time, waited for the customer to answer, and then proceeded to present the next question. The grounding manipulation mirrored the manipulation used in study 3. Immediately after the task, we assessed consumers' perceived brand intimacy (as in all previous studies; $\alpha_{\text{BrandIntimacy}} = 0.85$).

Pre-Test

We conducted a pre-test to assess the effectiveness of the experimental manipulation with a total of 50 participants (M_{Age}=38.64, SD_{Age}=12.19, 50% females) from the same population and using the same selection criteria as in the main study. Specifically, the objective of this pre-test was to assess whether the objective task was perceived as requiring less understanding from the brand compared to the subjective one. Participants were randomly assigned to the same two task conditions as in the main study (subjective vs. objective task). In each condition, they read a description of the task they would engage in. Specifically, in the subjective task condition, they read: "Imagine a recruiting advisory called AgileRecruit needs to deeply understand subjective information about potential

candidates' personality and soft skills to help you find suitable candidates to join your team. In order to find suitable candidates, AgileRecruit provides an automated system that asks you several questions about your candidate profile, such as what personality traits you are looking for, what matters most to you in a work relationship, and how your ideal interview would look like." In the objective task condition, participants read: "Imagine a recruiting advisory called AgileRecruit needs to collect objective information about potential candidates' profiles and hard skills to help you find suitable candidates to complete a business task. In order to record your information, AgileRecruit provides an automated system that asks you several questions about your candidate profile, such as their age, education level, and number of years of experience." Immediately after reading this scenario description, we assessed participants' perceived need for mutual understanding (sample item: "This task requires that the brand really understands me," $\alpha_{NMU} = 0.93$, web appendix 5.12). Supporting the effectiveness of the experimental manipulation, the objective task was perceived as requiring a significantly lower sense of mutual understanding compared to the subjective task ($M_{Objective} = 4.00$, $M_{Subjective} = 5.55$, $F(1, 48) = 16.990$, $p < .001$, η^2=0.26).

Manipulation Checks

To confirm the effectiveness of the grounding manipulation for both tasks, we measured participants' perceived degree of grounding based on the same five items as in study 3 ($\alpha_{Grounding} = 0.91$, web appendix 5.4). Supporting the effectiveness of our manipulation, the interface condition had a significant effect on the perception of grounding ($M_{Grounding_Present} = 5.37$, $M_{Grounding_Absent} = 4.06$, $F(3, 570) = 126.362$, $p < .001$, η^2=0.18). While the effect of the task was not statistically significant ($p = .08$) we observed a directional increase in perceived grounding in the subjective versus objective task ($M_{Objective} = 4.63$, $M_{Subjective} = 4.83$).

In addition, participants were asked to rate the objectivity of the task on a scale from 0 = not objective at all to 100 = extremely objective (Castelo et al. 2019). Supporting the effectiveness of our manipulation, the objective task was perceived as significantly more objective than the subjective task ($M_{Objective} = 69.7$, $M_{Subjective} = 59.8$, $F(3, 570) = 21.54$, $p < .001$, η^2=0.04) while neither grounding ($p = .922$) nor the interaction between the two experimental factors ($p = .202$) affected the perceived objectivity of the task.

Results

Brand Intimacy. Consistent with our previous findings, grounding had a positive main effect on brand intimacy ($M_{GroundingPresent} = 4.63$, $M_{GroundingAbsent} = 4.38$, $F(3, 570) = 4.455$, $p < .05$, η^2=0.01), and brand intimacy was significantly higher in the high-need-for-mutual-understanding condition as opposed to the low-need-for-mutual-understanding condition ($M_{HighNMU} = 4.63$, $M_{LowNMU} = 4.36$, $F(3, 570) = 4.860$, $p < .05$, η^2=0.01). Critically, in support of our hypothesis that the positive effect of grounding on brand intimacy is attenuated for tasks requiring low as opposed to high need for mutual understanding (hypothesis 6), we observed a significant interaction between both factors ($F(3, 570) = 10.165$, $p < .01$, η^2=0.02). Follow-up contrasts with Tukey HSD correction revealed a significant positive effect of grounding on brand intimacy in the task requiring high need for mutual understanding ($M_{GroundingPresent_HighNMU} = 4.94$, $M_{GroundingAbsent_HighNMU} = 4.32$, $p < .001$), while this effect was not significant in the task requiring low need for mutual understanding ($M_{GroundingPresent_LowNMU} = 4.29$, $M_{GroundingAbsent_LowNMU} = 4.44$, $p = .825$) (figure 10).

Discussion

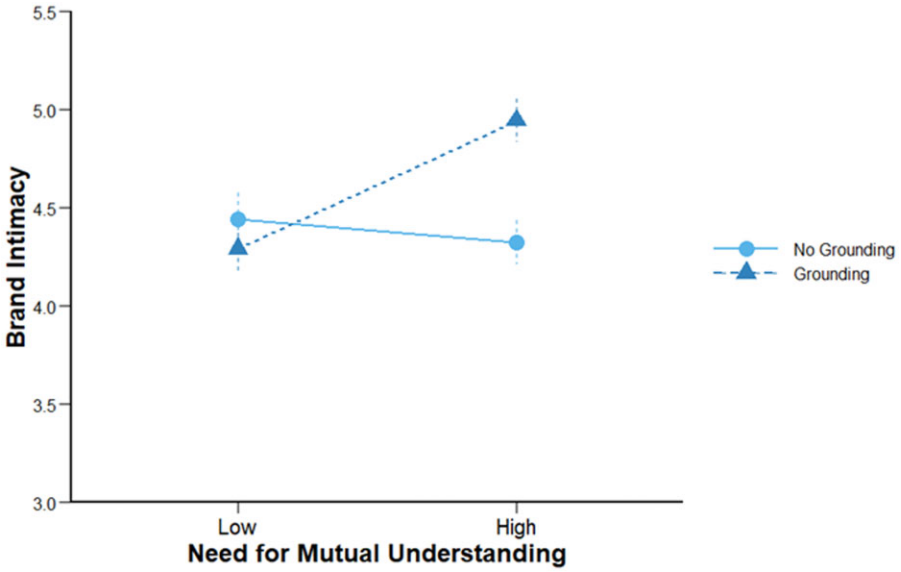
Study 4 provides evidence in support of the moderating role of need for mutual understanding (hypothesis 6). It demonstrates that grounding enhances brand intimacy in contexts that require a significant amount of mutual understanding, but not in contexts where that is not the case. As the nature of the content of the interaction (i.e., the extent to which it focuses on subjective vs. objective aspects) is not the only factor implying different levels of need for mutual understanding, we conducted a separate study to test hypothesis 6 in connection with another form of context-dependent need for mutual understanding. That study, which is reported in web appendix 7.2, provides a conceptual replication of study 4 in settings where a greater need for mutual understanding arises from greater decision complexity.

GENERAL DISCUSSION

Across four studies, we shed light on how fundamental properties of human dialogue (turn-taking, turn initiation, and grounding) shape how consumers perceive and interact with a brand's AI-based conversational interface. These properties induce greater perceptions of humanness, lead to more intimate consumer–brand relationships, and ultimately affect a series of downstream consequences for consumers (e.g., greater recommendation acceptance, stronger future commitment to the brand in terms of enhanced brand advocacy and loyalty). We find evidence for these effects across multiple product settings (from design and fashion contexts to automating beverage delivery). Finally, we demonstrate that a conversational interface's ability to promote a more intimate relationship with the brand is enhanced in situations that require higher (as opposed to lower) levels of mutual understanding.

FIGURE 10

EFFECT OF GROUNDING ON BRAND INTIMACY: MODERATING ROLE OF NEED FOR MUTUAL UNDERSTANDING



Theoretical Implications

The current work makes three novel contributions. First, the current research introduces a verbal embodiment framework to illuminate how conversational AI interactions alter consumers' technology-mediated brand experience (Parasuraman and Zinkhan 2002; Schmitt, Brakus, and Zarantonello 2015). We develop a verbal embodiment framework by integrating previously disconnected streams of research on humanizing brands and brand interactions (Holzwarth et al. 2006; MacInnis and Folkes 2017; Schroll, Schnurr, and Grewal 2018; Verhagen et al. 2014; Wang et al. 2007) with research on embodied systems (Bickmore and Cassell 2001; Cassell et al. 2000, 2001). We identify three critical conversational design characteristics (turn-taking, turn initiation, grounding) and demonstrate how their interplay facilitates the humanization of technology-mediated consumer–brand interactions. Our work advances our understanding of how conversational AI interactions, and their ability to mimic human behavior, influence how consumers feel and behave toward a brand. Thus, our work provides a new perspective on how branded conversational interfaces can shape, influence, and ultimately design consumer–brand relationships and responds to recent calls for more research on technology-mediated consumer–brand interactions (Melumad et al. 2020; Yadav and Pavlou 2014; Zierau et al. 2022).

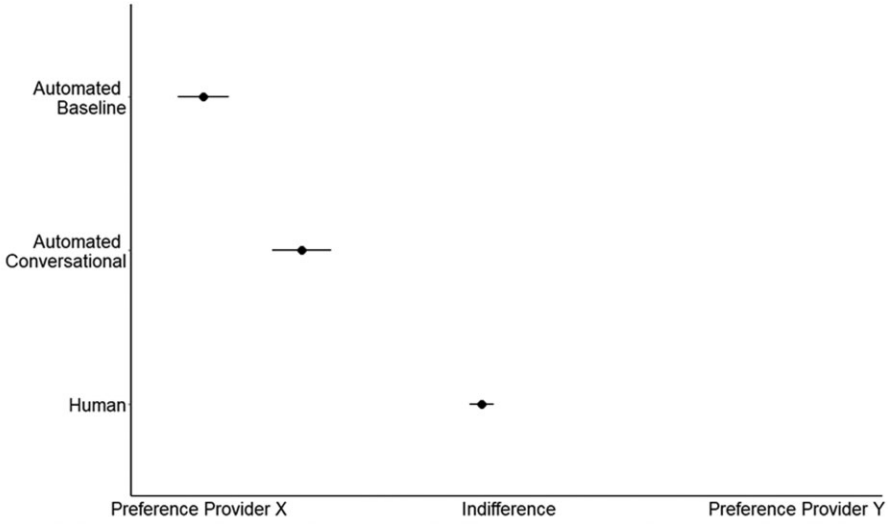
Second, the current research contributes to recent conceptual work on consumer–smart object relationships in marketing (Hoffman and Novak 2018; Novak and Hoffman 2018). The current work provides evidence that, even though a conversational interface acts merely as a medium to complete a task (such as ordering a cup of coffee), consumers tend to form a relationship with the medium and attribute meaning toward the emerging "assemblage" such that the relationship formed during the interaction with the medium extends toward the brand embodied through this medium (i.e., the chatbot). Thus, our work provides direct empirical evidence on how consumer–object assemblages alter consumer shopping experiences by creating a more intimate consumer–brand relationship.

Third, the current research also contributes to prior work on algorithm aversion and resistance toward AI. Our findings hint at the possibility that enhanced verbal embodiment can reduce consumers' aversion toward AI. In a recent work on resistance to medical AI, Longoni et al. (2019) suggest that the lack of perceived warmth may explain consumers' resistance to trust and to choose AI over a human to complete a task. The current research suggests that the intentional design of a conversational AI with enhanced turn-taking, turn initiation, and grounding can ascribe more human-like capacities to the AI and in turn promote more positive consumer–firm outcomes (from enhanced commitment or loyalty to more positively valenced firm perceptions).

We conducted a follow-up study to directly test the enhanced verbal embodiment of a conversational AI reduces consumers' resistance toward AI. Building on

FIGURE 11

PREFERENCE FOR HUMAN VERSUS AUTOMATED MEDICAL PROVIDER



NOTE.— Preference for provider X always resembled preference for a human dermatologist. Preference for provider Y resembled, depending on the experimental condition, consumers' preference for either a human dermatologist, an automated dermatologist, or an automated conversational interface.

Longoni et al.'s (2019) paradigm about skin cancer screenings in which consumers had to decide between a human or fully automated dermatologist for a screening service (study 3A, Longoni et al. 2019), we replicated the authors' two original conditions and one using a conversational interface employing turn-taking, tun initiation, and grounding (see web appendix 8 for details). As shown in figure 11, while consumers generally preferred a human over an automated provider ($M_{\text{Human}} = 3.85$, $M_{\text{AutomatedBaseline}} = 2.22$, $t = 5.439$, $p < .001$), consumer preference shifted significantly toward a conversational interface relative to the baseline of a simplistic automated medical provider ($M_{\text{AutomatedConversational}} = 2.80$, $t = 2.969$, $p < .01$, yet still significantly lower compared to human provider: $t = 5.439$, $p < .001$). Although these findings provide initial, preliminary evidence, they suggest that developing more relatable AI interactions through enhanced verbal embodiment holds the potential to significantly decrease consumers' resistance toward AI.

Practical Implications

To the best of our knowledge, this work is the first to provide a theory-driven framework to develop more relatable and effective AI-based conversational interfaces. Specifically, we provide a novel framework to enhance consumer–brand relationships by utilizing fundamental principles of human-to-human dialogue (turn-taking, turn initiation, and grounding). Our findings suggest that brands should spend more deliberate effort on optimizing the conversational design of how their interface takes turns during a conversation with consumers. At the most basic level, this includes making sure that the interface knows when to speak and when it is the consumers' turn to speak. We also advise brands to monitor how long individual turns are as this can affect perceptions of who dominates the conversation (Itakura 2001) and how intonations are employed to signal smooth iterations between turns and foster more enjoyable conversational AI interactions (Kannan 2022).

The insights on grounding in this research also highlight the context-specific sensitivity of grounding on consumer–brand relationships and that its effectiveness is reduced for tasks that require a low need for mutual understanding, such as more objective tasks or tasks that are less complex (see additional study in web appendix 7.2). This suggests that not all contexts require interfaces with enhanced verbal embodiment to maximize the perception of humanness. For example, in cases where the brand uses an interface to collect simple demographic information, or to provide a quick answer to a question, the brand may not benefit from adding grounding signals to their interface. However, when a brand uses a conversational interface to establish ongoing and more complex interactions with a consumer, the brand will positively benefit from adding grounding signals to the conversation as it will result in stronger consumer–brand relationships and more positive downstream outcomes for the brand. Thus, the current work enables practitioners to more rigorously and deliberately design a

brand's conversational interface with important design implications conditional on the situational affordances (such as reducing grounding interventions when the need for mutual understanding of the consumer and the brand is low).

Finally, harmonizing consumer–brand perceptions across touchpoints is a major issue in modern brand management. The current work highlights an opportunity to leverage conversational interfaces to "harmonize" this experience across consumer–brand touchpoints. Compared to deploying human service representatives (in both online and offline channels), conversational interfaces are a cost-effective strategy in marketers' brand management toolbox, providing a high level of control as well as consistency in the way brands engage with their customers.

Conclusion

Conversational interfaces are more than cost-effective tools to automate consumer–firm interactions. The current work shows that AI-based conversational interfaces provide a powerful branding ecosystem that delivers highly scalable one-to-one consumer–brand experiences, nurtures consumer–brand relationships, and systematically influences a broad range of behavioral brand outcomes (from greater recommendation acceptance to increased loyalty). The verbal embodiment framework introduced and tested here advances our understanding of how AI-based conversational interfaces deepen consumer–brand relationships in an increasingly AI-driven marketplace.

DATA COLLECTION INFORMATION

The first and second authors managed the setup and data collection for study 1 via FlowXO (chatbot platform) and Unipark (experiment and survey platform) using MTurk participants in 2018. The second author led the data analysis for study 1 and results were discussed among all three authors. The first and second authors managed the setup and data collection for study 2 via FlowXO (chatbot platform) and Unipark (experimental and survey platform) using Prolific Academic in 2020. The data were analyzed by the first and second authors and the results were discussed jointly among all three authors. The first author managed the setup and data collection for studies 3 and 4 via FlowXO (chatbot platform) and Unipark (experimental and survey platform) using Prolific Academic in 2021 and 2022. Data were discussed jointly and analyzed by the first author. The research design, analyses, and final results of all studies were discussed among all three authors. The full datasets and processing scripts used in studies 1–4 are available on the Open Science Framework: https://osf.io/ejvhu/?view_only=181aae699eff42ae877054fe9e846ffc.

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